B1 sparse hybrid with calibration CV: Methodology and Evidence

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Status. Post-run methodology reconstruction grounded in archived source and reports. This document is not an executed historical notebook.

Research question

Check whether the strong sparse hybrid depended too much on one calibration split's fitted thresholds. The cross-validation route limits optimistic threshold fitting without using official dev labels for selection.

Architecture and mechanism

The same 0.9 sparse/0.1 B1 blend is evaluated with five-fold cross-validation inside the official-train calibration subset. Class thresholds shrink toward a global prior according to support; the final thresholds are refit on all calibration rows.

Measured result

Scored submission	1016805
Codabench macro-F1	0.3944
Archived local metric	0.3933 official-dev macro-F1

The report records calibration CV macro-F1 0.4282, official-dev macro-F1 0.3933, and Codabench 0.3944. The latter is effectively tied with the fixed hybrid and does not establish a hidden-score gain.

Limitations and next check

Five-fold calibration still has variance for rare labels; selection is not a substitute for a larger independent development set. Local, pilot, and hidden metrics are deliberately kept distinct. A single scored run does not establish a causal architecture effect or statistical significance.

Source and reproducibility

Original source modules: phase5_hybrid_calibration_cv.py, phase5_prepare_hybrid_probs.py, phase5_b1_train.py, run_logistic.py. SHA-256 hashes, the original report, and the prediction ZIP receipt are in the companion provenance file. The explanatory notebook was written after the historical run; original execution output is never implied. Reproduction requires authorized course data, the Phase 1 parent weights/tokenizer, and the dependencies in the archived source.

Documentation disclosure

AI assistance was used to organize the archived evidence and draft this explanatory document. All numerical claims above come from saved run records or the provided finished-submission screenshot; missing evidence is labelled as such.